MARIA GARCIA-OSIPENKO

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EDUCATION

Arizona State University	Ph.D. in Economics	Expected May 2026
Arizona State University	M.S. in Economics	December 2022
Siberian Federal University	B.A. in Economics	June 2020
University of Minnesota	Exchange Student, Economics	2017-2018

RESEARCH FIELDS

Environmental Economics, Energy Economics, Transportation Economics, Industrial Organization

WORKING PAPERS

"Technology Complementarities and Subsidy Policy: Evidence from Electric Vehicle and Solar Panel Adoption" (*Job Market Paper*)
Revise and Resubmit, Journal of the Association of Environmental and Resource Economists
Government policies target air pollution and climate change by incentivizing adoption of electric vehicles (EVs) and/or residential solar panels (PVs). Knowledge of whether these goods are complements or substitutes can be used to design policies that target environmental externalities more efficiently. I use California household-level data to estimate a structural multi-product demand model. I find that consumers view PVs and EVs as complements, with the degree of complementarity varying with vehicle size and income. Counterfactual experiments reveal that complementarity significantly increases bundled EV-PV purchases. This complementarity can be leveraged to design policies that achieve emission targets at lower cost.

"Endogenous Rigidities and Capital Misallocation: Evidence from Containerships" *with Nicholas Vreugdenhil and Nahim Bin Zahur*
Revise and Resubmit, Journal of Political Economy
We investigate how endogenous rigidities inhibit physical capital reallocation. We focus on the role of contract duration - a classic example of an adjustment rigidity. We argue that when agents sign longer contracts in booms when markets are thin, they generate a contracting externality which further amplifies thinness and impedes the adjustment of markets to shocks. We develop a framework with booms and busts where agents search and choose match duration. Applying the framework to the containership leasing market, we find substantial misallocation from endogenous rigidities, particularly in the transition after a crash. We also quantify implications for designing industrial policy.

"Optimal Second-best Menu Design: Evidence from Residential Electricity Plans" *with Nicolai Kuminoff, Spencer Perry, and Nicholas Vreugdenhil*
Under Review
Utilities increasingly sell electricity using complex menus of time-constant and time-varying price schedules. We study how to design such a menu to maximize social welfare in a second-best environment where the marginal private and external costs of generating electricity vary over time, institutional constraints prevent mandating time-varying pricing, and consumer behavior is distorted by frictions. We develop a model of plan choice, consumption, and intertemporal substitution with time-varying marginal social costs, and estimate it using administrative data from a large utility. We provide evidence of substantial intertemporal substitution in response to time-varying price incentives, and selection across plans based on multidimensional heterogeneity. While the current menu's time-varying plans substantially shift consumption from high-price to low-price hours, we find that they reduce social welfare. This loss is mitigated by information frictions. We show how to redesign the menu to simultaneously improve outcomes for consumers, the utility, and the environment.

<u>"Optimizing Electricity Rates: Evidence from Commercial and Industrial Firms"</u>

Commercial and industrial firms account for over half of total electricity demand in many U.S. utilities, yet little is known about how they choose among electricity rate options or adjust consumption in response to prices. Although system costs vary substantially throughout the day, most firms remain on time-invariant price plans. Time-of-use (TOU) pricing introduces temporal price variation to better align demand with system costs, but its effectiveness depends on design. Using administrative data from a large U.S. utility, I document that firms switching to TOU pricing reduce peak-period demand, with significant heterogeneity across industries. I then develop and estimate a structural model of rate selection and hourly electricity consumption that incorporates operational flexibility, price responsiveness, and adjustment frictions. Identification leverages variation from switching plans. The estimated framework enables counterfactual analyses of alternative pricing menus and their implications for customer surplus, system costs, and environmental outcomes.

PUBLICATIONS

"Green Economy as a Labor Productivity Factor in the Manufacturing Industry of European Union Countries" with Vladislav N. Rutskiy**,** *Financial Journal***,** 2020

RESEARCH EXPERIENCE

Arizona State University

August 2023 - Present: Research Assistant for Nicolai Kuminoff

May 2022 - Present: Research Assistant for Nicholas Vreugdenhil

Siberian Federal University

Dec 2016 - May 2017: Research Assistant for Vladislav N. Rutskiy and N N Tarun Chakravorty

ACADEMIC PRESENTATIONS

2025 Environmental and Energy Economics Workshop at the University of Arizona

2024 SWEEEP at Georgia Tech University, Energy Camp at UC Berkeley, AERE Summer Conference

2023 NCSU Camp Resources, Berkeley/Sloan Summer School in Environmental and Energy Economics, AERE Sessions at the Western Economic Association Conference, Arizona Workshop on Environment, Natural Resources, and Energy Economics

TEACHING EXPERIENCE

Arizona State University, Instructor

Summer 2023: Microeconomic Principles

Arizona State University, Teaching Assistant

Spring 2023: Public Economics

Fall 2022: Industrial Organization and Competition Policy

Fall 2021, Spring 2022: Environmental Economics

Spring 2022: Business Statistics

Fall 2021: Macroeconomic Principles

AWARDS

2023 Rondthaler Emeritus Award "Most Deserving Graduate Student in the Program", ASU Economics Dept

2023 "Best Progress Towards Dissertation", ASU Economics Dept

2023 Travel Grant Award, ASU Graduate and Professional Student Association

GRANTS

Predicting Commercial and Industrial Electricity Load. PhD Student Investigator. 2024-2025. Salt River Project, $68,866 (*PI: Nicolai V. Kuminoff, Co-PI: Nicholas Vreugdenhil*)

Predicting Residential Price Plan Enrollment and Energy Use. PhD Student Investigator. 2023-2024. Salt River Project, $50,042. (*PI: Nicolai V. Kuminoff, Co-PI: Nicholas Vreugdenhil*)

ADDITIONAL INFORMATION

Languages: English (fluent), Russian (native)
Programming: Stata, Python, MATLAB, LaTeX, Git, GitHub
Citizenship/Work Authorization: U.S. Permanent Resident (green card)

REFERENCES

Nicolai V. Kuminoff (Co-Chair)
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PLACEMENT CONTACTS

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